

KERALA AGRICULTURAL UNIVERSITY

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OIL SEEDS

COCONUT (*Cocos nucifera*)

Coconut requires an equatorial climate with high humidity. The ideal mean annual temperature is 27°C with 5-7° diurnal variation. The palm does not withstand prolonged spells of extreme variations. A well-distributed rainfall of 1300-2300 mm per annum is preferred.

Coconut is grown in different soil types such as laterite, coastal sandy, alluvial, and also in reclaimed soils of the marshy lowlands. It tolerates salinity and a wide range of pH (from 5.0-8.0).

The cultural practices have to be adopted to suit the varying climatic and soil conditions.

Selection of site

Select sites with deep (not less than 1.5 m depth) well drained soil. Avoid shallow soils with underlying hard rock, low-lying areas subject to water stagnation and heavy clayey soils.

Cultivars

1. West Coast Tall (WCT)
2. Lakshadweep Ordinary (Chandrakalpa)
3. Philippines Ordinary (Kerachandra)
4. Andaman Ordinary
5. Java
6. Cochin China
7. Kappadam
8. Komadan
9. Kerasagara
10. Kalparaksha
11. Kalpadhenu
12. Kalpaprabhitha
13. Kalpamithra

Hybrids

1. Lakshaganga (Lakshadweep Ordinary x Gangabondam)
2. Anandaganga (Andaman Ordinary x Gangabondam)
3. Keraganga (West Coast Tall x Gangabondam)
4. Kerasankara (West Coast Tall x Chowghat Orange Dwarf)
5. Chandrasankara (Chowghat Orange Dwarf x West Coast Tall)
6. Kerasree (West Coast Tall x Malayan Yellow Dwarf)
7. Kerasoubaghya (WCT x SSA)
8. Chowghat Green Dwarf x West Coast Tall
9. Chandralaksha (Lakshadweep Ordinary x Chowghat Orange Dwarf)

Tender nut variety: Chowghat Orange Dwarf

Note:

- (1) Hybrids Anandaganga, Keraganga and Kerasankara are recommended for general cultivation both under rainfed and irrigated conditions.
- (2) Other hybrids especially Chandrasankara are recommended for ideal situations and where good management practices are adopted.
- (3) Since the performance of Chandrasankara is markedly superior to that of WCT in root (wilt) affected areas, cultivation of Chandrasankara and Kalparaksha are preferred in such areas.
- (4) Chandralaksha, Lakshaganga, Chandrakalpa and Kalpadhenu are recommended for cultivation under drought prone areas.
- (5) Kalparaksha is recommended for tender coconut. Kalpaprabhitha is a dual purpose variety (copra and tender coconut).
- (6) Kalpamithra is recommended for rainfed conditions.

Selection of mother palms

Select mother palms having the following characters:

1. Regular bearing habit and yielding not less than 80 nuts / annum.
2. Age 20 years or more (5 years after reaching full bearing capacity). If the mother palms are the progeny of elite planting material and gives consistently higher yields for a period of not less than 6 years, seed nuts can be

collected from such palms. There is no need for insisting 20 years as minimum age for mother palms in such conditions.

3. More than 30 fully opened leaves with short strong petioles and wide leaf base firmly attached to the stem.
4. Bearing at least 12 bunches of nuts with strong bunch stalks.
5. Bearing nuts of medium size and oblong shape.
6. Husked nuts should weigh not less than 600 g.
7. Mean copra content of 150 g per nut or more.

Avoid palms which (i) have long, thin and pendulous inflorescence stalks (ii) produce long, narrow, small sized or barren nuts (iii) show shedding of immature nuts in large numbers and (iv) are grown under favourable environmental conditions.

Collection and storage of seed nuts

Collect mature nuts (above 11 month old) during the period from December to May. Lowering of bunches by means of ropes may be done when the palms are tall and ground is hard. Discard nuts, which show improper development or other undesirable features. Store seeds in shade for a minimum period of 60 days prior to sowing in nursery. For storing, arrange the seed nuts with the stalk-end up over an 8 cm layer of sand in a shed and cover with sand to prevent drying of nut water. Up to five layers of nuts can be arranged one over the other. The nuts can also be stored in plots, provided the soil is sandy and the ground is sufficiently shaded. In the case of nuts harvested in May, heap them in partial shade, till husk is well dried and then sow them in the nursery.

Selection and preparation of site for nursery

Nursery sites should be well drained with light textured soil and with adequate but not too much shade. In open areas, provide shade during summer. Prepare beds of 1.5 m width and of convenient length with 75 cm space between beds. In areas where drainage is poor, prepare raised beds.

Before planting, examine seed nuts and discard those without nut water and rotten kernels. Sow the nuts in the nursery after commencement of southwest monsoon during May-June.

Spacing of nuts

Plant the seed nuts at a spacing of 30 cm (between rows) x 30 cm (between nuts) with four or five rows per bed.

Method of planting seed nuts

Plant the seed nuts in the beds in trenches 25-30 cm deep and cover with soil so that top portion of husk alone is visible. The nuts may be planted either horizontally with the widest of the segments at the top or vertically with stalk-end up. Vertical planting is preferable on account of convenience in transporting and lesser risk of seedling injury.

Care and management of nursery

Provide protective fencing to the nursery if it is located in open area. If the soil is sandy, provide mulching immediately after the cessation of monsoon rain. Irrigate the nursery once in two days during summer months. Keep the nursery beds free of weeds by periodic weeding. If termite is noticed, remove soil in the affected area upto a depth of about 15 cm and dust soil and nuts with carbaryl or chlorpyrifos. Repeat if attack persists. Periodically spray the plants with 1 per cent bordeaux mixture or any other copper fungicide to prevent fungal infection.

Selection of seedlings

Remove seed nuts, which do not germinate within 6 months after sowing as well as those with dead sprouts. Select only good quality seedlings (9-12 months old) by a rigorous selection based on the following characteristics.

1. Early germination, rapid growth and seedling vigour.
2. Six to eight leaves for 10-12 months old seedlings and at least four leaves for 9 months old seedlings.
3. Collar girth of 10-12 cm.
4. Early splitting of leaves.

Note: The recovery of quality seedlings will be about 60-65 per cent. Since early germination is one of the criteria for the selection of seedlings, the storing and sowing of seed nuts should be in lots rather than in a staggered manner.

Removal of seedlings

Remove seedlings from the nursery by lifting with spade and cutting the roots. Keep the seedlings in shade and do not expose to sun. Plant seedlings as early as possible after removal from nursery. Never allow lifting the seedlings from the soil by pulling the leaves or stem.

Preparation of land and planting of seedlings

The nature of preparation of land before planting depends upon topography of land, soil type and other environmental factors. On slopes and in areas of undulating terrain, prepare the land by contour terracing or bunding. In low-lying areas and rice fields, form mounds to a height of at least 1 m above water level. In reclaimed kayal areas, planting can be done on the field bunds.

The size of pits for planting would depend upon soil types and water table. In loamy soils with low water table, pit size of 1 m x 1 m x 1 m is recommended. In laterite soils with underlying rock, take larger pits of size 1.2 m x 1.2 m x 1.2 m. In sandy soils, the size of pits may be 0.75 m x 0.75 m x 0.75 m. The pits may be filled up with top soil to a height 60 cm below the ground level.

In low lying lands, take shallow pits and as the plant grows, raise the ground level by adding silt and sand so as to cover the entire bole of the palm. The same procedure can be adopted when planting is done on mounds or bunds. Burial of two layers of husks in the pits will be useful for moisture conservation. The husk is to be buried in layers with concave surface facing upwards. After arranging each layer, sprinkle carbaryl 10 per cent dust on the husk to prevent colonization by termites.

Note: In lateritic areas, common salt @ 2 kg per pit may be applied on the pit to improve soil conditions. Common salt is to be applied about six months prior to planting.

Spacing

Spacing depends upon the planting system, soil type etc. In general, the following spacings are recommended under different systems in sandy and laterite soils. In lateritic gravelly soils, under rainfed conditions of north Kerala, a closer spacing to accommodate 250 palms per ha is more economical.

Table 18. Spacing for coconut

Planting system	Spacing	Approximate number of plants/ha ⁻¹
Triangular	7.6 m	198
Square	7.6m to 9 m	170-120
Single hedge	5 m in the rows 9 m between the rows	220
Double hedge	5m x 5 m in rows 9 m between pairs of rows	280

In the hedge system of planting, the rows should be aligned in north-south direction and the seedlings planted as in the triangular system.

Time of planting

Planting the seedlings during May, with the onset of pre-monsoon rains is ideal. Under assured irrigation, planting can be done during April also. In low-lying areas, plant the seedlings in September after the cessation of heavy rains.

Shading and irrigation

For the first two years from planting, irrigate @ 45 litres of water per seedling, once in 4 days, during dry summer months. Provide adequate shade to the transplanted seedlings.

Manuring young palms

For the first three years after planting under rainfed conditions, apply fertilizers in two split doses at the rates shown in Table 19. Fertilizer requirement of adult palms is given in Table 20.

Table 19. Fertilizer requirement of young palms in relation to that of adult palms

Time after planting	Time of application	
	April-June	Sept-Oct.
	(Proportion of adults palm dose)	
3 months (1/10th of full dose)		1/10
1 year (1/3rd of full dose)	1/9	2/9
2 year (2/3rd of full dose)	2/9	4/9
3 year onwards (full dose)	3/9	6/9

Note: Under irrigated conditions, the fertilizers can be applied in 3-4 equal split doses. In the case of low lying areas, apply fertilizer after water table recedes in one single dose or in two split doses as conditions permit. In all types of soils that are low in organic matter content (except reclaimed clayey soils and alluvial soils), apply organic matter @ of 15-25 kg per palm per year during June-July from the second year of planting.

Weeding and interculture

Keep the pits free of weeds by periodical weeding. Remove the soil covering the collar of seedlings. As the seedlings grow fill up the pits gradually by cutting the sides. Proper intercultivation provides control of weeds and

creates soil mulch. Any tillage system (ploughing, digging, raking or forming mounds) that provides soil mulch and control weeds may be followed depending upon local conditions. For laterite, sandy and red sandy loam soils give two ploughings or diggings in May-June and September-October and one raking in January. In areas where surface run off is more, form mounds in September-October and level them in November-December.

Drought management in coconut gardens

Coconut produces nuts round the year. Therefore, adequate supply of water is essential for its proper growth. Soil moisture is essential for the absorption of nutrients by roots. Moisture stress leads to stunted growth, drooping of leaves, immature nut fall and decreased yield. Importance may be given on the following aspects so as to ward off stress:

1. Husk burial for moisture conservation

Burying of fresh or dried coconut husk around the palm is a desirable practice particularly for moisture retention. The husk can be buried either in linear trenches taken 3 m away from the trunk between rows of palms or in circular trenches taken around the palm at a distance of 2 m from the trunk. The trenches may be of 0.5 m width and depth. The husks are to be placed in layers with concave surface facing upwards and covered with soil. The beneficial effect of husk burial will last for about 5-7 years. Instead of husk, coir pith can be buried @ 25 kg per palm per year.

2. Mulching

Mulching is an effective method of conserving soil moisture. Mulch the coconut basins with green / dry leaves at the close of northeast monsoon (October-November). Mulching also adds organic matter to the soil and reduces the soil temperature. Do not disturb soil in the coconut garden during summer months. In level lands, during rainy seasons excess water may be conserved in small trenches dug out in the plantation. In sloppy areas, land may be terraced and trenches dug across. This will facilitate maximum percolation of rainwater and water conservation. For moisture conservation, lowermost 3-5 leaves may be cut and removed. Provide adequate shade for the transplanted seedlings for 1-2 years. To minimize the sun scorch on the trunk, application of lime solution on the trunk up to a height of 2-3 m at the start of the summer season is recommended.

3. Green manure and cover crops

Green manure and cover crops recommended for cultivation in coconut gardens are:

(a) Green manure crops: *Crotalaria juncea* (sunhemp), *Tephrosia purpurea* (kolinji), *Indigofera hirsuta*, *Pueraria phaseoloides*.

(b) Cover crops: *Calapagonium muconoides*, *Mimosa invisa*, *Stylosanthes gracilis*

(c) Shade-cum-green manure shrub: *Tephrosia candida*

Sow cowpea seeds more towards the periphery of basins taken at a radius of 2.0m from the base of the palm for green manure during April-May with the onset of pre-monsoon rains. When a few plants start flowering, uproot the entire plants and incorporate into the soil during August-September and cover the basins with soil.

Sow green manure and cover crop seeds during April-May with the onset of pre-monsoon rains. The green manure crops should be ploughed in and incorporated into the soil during August-September. This will increase the water holding capacity of soil. *Calapagonium* can be grown either as green manure or cover crop. *Tephrosia* is especially suited for planting around seedling pits for summer shade and as a source of green manure in the rainy season.

Manuring of adult palms

The nutrient dosages recommended for adult palms are given in Table 20.

Table 20. Fertilizer recommendation for coconut

		Quantity,kg/palm/annum		
		N	P ₂ O ₅	K ₂ O
1	General recommendation	0.34	0.17	0.68
	(a) Average management	0.50	0.32	1.20
	(b) Good management			
2	For reclaimed clayey soils (as in Kuttanad)	0.25	0.35	0.90
3	Red loam soils (southern Kerala)	0.68	0.23	0.90
4	Hybrids & high yielding palms	1.00	0.50	2.00
	(a) For irrigated areas	0.50	0.32	1.20
	(b) For rain fed conditions			

1. Under irrigated conditions, fertilizers can be applied in 3-4 equal split doses.

2. In the case of low-lying areas, apply fertilizers in one single dose after water table recedes or in two split doses as conditions permit.

3. The application of organic materials such as forest leaves, cattle manure, coir dust or coconut shredding at 10 kg per pit in the first three years and 15-25 kg thereafter will be useful to obtain better establishment of coconut palms in sandy soils and in coastal situations.

4. In situations where the available P_2O_5 status of the soil is more than 10 ppm, application of phosphatic fertilizers can be skipped for a few years until the status of P_2O_5 reaches 10 ppm.
5. For sandy and sandy loams of Onattukara and similar situations and also for hybrid palms grown in root (wilt) affected areas, apply 500 g N + 300 g P_2O_5 + 1000 g K_2O along with 500 g $MgSO_4$ per palm per year.
6. Application of $MgSO_4$ to coconut palms earlier confined to root wilt affected areas is recommended for the whole state (*ad hoc* recommendation).
7. The N: P_2O_5 : K_2O recommendation given for high yielding palms is, in general, sufficient for palms yielding upto 100 nuts per year. For palms yielding more than 100 nuts per year, an additional dose of 10 g N, 5 g P_2O_5 and 15 g K_2O may be supplied for every nut exceeding 100 nuts (*ad hoc* recommendation).
8. In laterite soils, 50 per cent of the K_2O requirement of coconut can be substituted by Na_2O supplied in the form of sodium chloride.

Time, frequency and method of fertilizer application

Under rainfed conditions, apply fertilizers in two split doses, 1/3 at the time of early southwest monsoon showers in April-June and 2/3 in September-October.

Under irrigated conditions, apply fertilizers in three or four equal doses in April-May, August-September, December and February-March.

Apply lime or dolomite during April-May, magnesium sulphate during August-September and organic matter during June-July. For an adult palm 1.0 kg dolomite or 1.0 kg lime + 0.5 kg $MgSO_4$ is required per annum.

Apply fertilizers and manures in circular basins at a radius of 2.0 m from the base of the palm and 10 cm deep, opened after the onset of southwest monsoon. Split doses can be applied with irrigation water in summer months.

Recycling of palm waste

Recycling of palm waste is very much beneficial especially for maintaining the availability status of micronutrients and trace elements. Palm wastes like coconut leaves, crown waste, dried spathes, husk etc. may be deposited in a small trench of convenient length, 0.5 m to 0.75 m wide and 0.3 to 0.5 m deep at a distance of 2-2.5 m away from the base of the trunk. Fill up this trench with the palm wastes along one side of the palm (north) in one year, opposite side (south) in the next year, east in the third year and so on. This practice of organic recycling of waste has been found to improve the growth and productivity of the palms.

Intercropping and mixed cropping

Schedules for inter/mixed cropping may be drawn up based on the canopy size, age and spacing of palms. In general, palms in the age group of 8-25 years are not suitable for inter and mixed cropping. However, cereals and tapioca are recommended as intercrops in young coconut plantation upto 3-4 years. Since ginger and turmeric are shade tolerant crops with shallow roots, they can be intercropped in coconut garden even in the age group of 15-25 years. It ensures better land utilization, solar energy harvesting, efficient water use, utilization of soil nutrient resources, more returns and an insurance against crop failure. Under conditions of wider spacing i.e. beyond 7.6 m, intercropping is possible irrespective of the age of the palms.

The following crops are recommended as intercrops.

Cereals: Rice, maize

Legumes and pulses: Groundnut, horse gram, cowpea.

Tubers: Tapioca, sweet potato, yams, colocasia

Spices and condiments: Ginger, turmeric, chilly, pepper, nutmeg, cinnamon, clove

Fruit plants: Banana, pineapple, papaya. (Banana variety Palayankodan is recommended in the reclaimed soils of Kuttanad. Three suckers per clump have to be retained).

Beverage crop: Cocoa

Fodder grasses: Hybrid Napier, guinea grass

In all cases, separate application of adequate fertilizers and manures to the individual crop is essential.

Casuarina (*Casuarina equisetifolia*) may be planted as single row intercrop, between rows of coconut during the first year of planting coconut seedlings in pits of size 30 cm x 30 cm x 30 cm spacing of 2m. The casuarina poles will be ready for felling/harvesting within 5-6 years of planting.

Crop cafeteria for multiple cropping in coconut garden

Perennials: Cocoa, nutmeg, pepper, clove, lemongrass and cinnamon.

Annuals:

(a) Kharif: Rice, maize, groundnut, ginger, turmeric, chilli, yams, colocasia, red gram, vegetables, sweet potato, tapioca, banana, pineapple, papaya and fodder grass.

(b) Rabi: Sesame, horse gram, red gram, vegetables, cowpea, sweet potato and banana.

(c) Summer: Vegetables

Irrigation

Irrigate the palms during summer months in basins around palms as shown below:

Table 21. Irrigation requirement of coconut

Parameters	Soil texture			
	Sandy	Sandy loam	Loam	Silty clay
Available soil moisture (cm/m)	8	12	17	21
Quantity of water / irrigation / palm in litres in a basin of 1.8 m radius	600	900	1300	1600
Frequency of irrigation (days)				
All areas in Kerala except north eastern portion of Thrissur and Palakkad districts	3-4	5	7-8	9
North eastern portion of Thrissur and Palakkad districts	2-3	3-4	5-6	6-7

Note: In coastal sandy soils, seawater can be used for irrigation. In irrigated gardens, interruption of irrigation would lead to serious set back in yield and general condition of palms. Hence, when once started, irrigation should be continued regularly and systematically. In sandy loam soil, irrigating the crop with 500 litres of water through basin taken at 1.5 m radius at CPE value of 50 mm (approximate interval of 15 days) is most economical. Do not irrigate seedlings and very young palms with seawater.

Drip irrigation

In the traditional system of irrigation followed in coconut gardens such as flood irrigation, basin irrigation etc. irrigation efficiency is only 30 to 50 per cent due to considerable wastage of water. In addition, cost on inputs like labour and energy in adopting these systems are high. Scarcity of water and increasing cost of labour and energy are deterrents in adopting these traditional irrigation systems. Under these circumstances, drip irrigation is the most suitable system of irrigation to coconut. Some of the major advantages of drip irrigation are: it saves water, enhances plant growth and yield, saves energy and labour, most suited for soils having low water holding capacity and undulating terrain, reduces weed growth and improves efficiency of fertilizers. For coconut, generally, three to four drippers are given per palm. The water requirement for an adult palm is 40 to 50 litres per day.

D x T hybrid production

The following guidelines are suggested for large scale production of D x T hybrid seedlings. Assisted pollination should be done to get maximum hybrid nut production. As far as possible use prepotent palms as parents in the hybridization programmes.

Selection of mother palms

Palms with the following phenotypic character combination may be selected for hybridization work.

1. Nuts without ridges and having yellow, orange or red colour.
2. Palms with overlapping female and male phases.
3. Small crown and canopy compared to that of tall palm.
4. Narrow stem without any bulging at the base with close leaf scars.

Hybridization

1. Use mixed pollen from identified tall palms.
2. Emasculate the inflorescence by cutting the male flowers with scissors and stripping if necessary within 5-7 days of opening the spathe.
3. Cloth bags made of very close mesh should be used for covering the inflorescence.
4. Hairy caterpillar larvae cause serious damage by boring into the female flowers and developing buttons through stigmatic ends. The damage is more serious under bagged conditions. Spraying 0.15 per cent carbaryl suspension mixed with 1 per cent fish oil soap or sandovit ensures protection from the larvae. Spraying may be done prior to pre-pollination bagging.
5. Dusting of pollen-talc mixture in 1:9 proportion using pollen dispensers is recommended.
6. Assisted pollination for at least 3-5 days on each inflorescence till last female flower becomes receptive and fully pollinated.
7. Remove bags after the seventh day of pollination of the last female flower.

Nursery

The nuts should be harvested before it is tree-ripe and sown immediately in the bed without storage. Nursery beds should be mulched or shaded and watered regularly and adequately.

Button shedding

The shedding of buttons in the coconut is attributed to the following reasons.

1. Pathological conditions
2. Attack of insect pests
3. Nutritional deficiencies
4. Soil and climatic variations
5. Defects in pollination and fertilization
6. Structural defects in the flower
7. Abortion of embryos
8. Limited capacity of the tree to bear fruits
9. Unfavourable conditions such as deficit of moisture, waterlogging and lack of aeration. The causes of button shedding may be identified and appropriate remedial measures adopted.

Plant protection

Pests

Rhinoceros beetle (*Oryctes rhinoceros*)

Symptoms

The adult beetle bores into the unopened fronds and spathes. The attacked frond when fully opened shows characteristic triangular cuts.

Management

1. Provide field sanitation to prevent breeding of beetles.
2. Hook out the beetles from the attacked palms by using beetle hook.
- 3 The topmost three leaf axils around the spindle may be filled with any of the following mixtures as a prophylactic measure:
 - (a) Application of 250g neem cake or marotti (*Hydnocarpus wightiana*) cake mixed with equal volume of sand in the innermost 2-3 leaf axils. This treatment is to be done twice, ie, during April-May before the onset of south-west monsoon and during September-October after the south-west monsoon.
 - (b) Naphthalene balls 12.0 g (approx. 4 nos.) in the innermost 2 leaf axils and covered with fine sand, once in 45 days.
 - (c) Incorporation of *Clerodendron infortunatum* @ 10 per cent w/w basis in the cowdung pit and/or manure pits
4. Treat manure pits and other possible breeding sites with 0.01 per cent carbaryl on w/w basis. Treatment will have to be repeated every three months.
5. Release *Baculovirus oryctes* infected adults @ 10-15 ha⁻¹ to bring down the pest population.
6. Inoculation of breeding sites with entomopathogenic fungus *Metarrhizium anisopliae* (@ 5 x 10¹¹ spores / m³) var major causes mortality to the grubs.

Red palm weevil

(*Rhynchophorus ferrugineus*)

Symptoms

The diagnostic symptoms are the presence of holes on the stem, oozing out of a viscous brown fluid and extrusion of chewed up fibrous matter through the hole, longitudinal splitting of leaf base and wilting of central shoot. Sometimes the gnawing sound produced by the feeding grubs inside will also be audible.

Management

1. Field sanitation should be given prime importance.
2. Avoid making steps or any other injury on the tree trunks to reduce the loci of infestation.
3. Leaf axil filling as suggested in the case of rhinoceros beetle will be useful against the red palm weevil also.
4. When green leaves are cut from the palms, stumps of not less than 120 cm may be left on the trees in order to prevent successful inward movement of the grubs through the cut end.
5. In attacked palms, observe for the bore- holes and seal them except the top most one. Through the top most hole, pour 1 per cent carbaryl or 0.15 per cent trichlorophon suspension @ one litre per palm, using a funnel.
6. When the pest infestation is through the crown, clean the crown and slowly pour the insecticidal suspension.
7. As an alternative, apply 1 per cent DDVP as a curative measure.
8. Coconut log traps with fermenting toddy or pineapple or sugarcane activated with yeast or molasses can be set in coconut plantation to attract and trap the free floating population of red palm weevil. Incorporate any of the insecticide to each trap to kill the weevils trapped.
9. Use of pheromone trap for attracting and killing adult weevils @ one trap per 2 ha.

Leaf eating caterpillar (*Opisina arenosella*)

Symptoms

The caterpillar feeds on green matter from the lower leaf surface, remaining within galleries of silk and frass. The attack will be severe during summer months from January-May.

Management

1. As a prophylactic measure, the first affected leaves may be cut and burnt during the beginning of the summer season.
2. Arrange for the release of larval / pupal parasitoids, *Goniozus nephantidis*, *Elasmus nephantidis* (brown species) and *Brachymeria nosatoi*.
3. Trunk release of *G.nephantidis* @ 10 nos/plam (4 to 6 releases) is effective for the management of leaf eating caterpillar.
4. When infestation is very severe and if the biocontrol is not likely to be effective, spray the undersurface of the fronds with dichlorvos 0.05 per cent, malathion 0.1 per cent, quinalphos 0.05 per cent, or phosalone 0.07 per cent.

Note: Application of the insecticides should be followed by liberation of larval and pupal parasites from the 21st day.

Cockchafer beetle

(*Leucopholis coneophora*)

Symptoms

The soil inhabiting white grubs cause damage to the roots of coconut palm. The attack is common in sandy tracts. The infested palms turn pale yellow and there will be considerable reduction in yield.

Management

1. Collection and destruction of adults during the monsoon period from adjacent vegetation (in the evening).
2. Plough or dig the infested soil synchro-nizing with pre-monsoon showers.
3. Drench the soil with chlorpyrifos 0.04 per cent suspension. The treatment should be given twice, first during April-May after the receipt of pre-monsoon showers and second during the month of September.

Note: Wherever possible, light traps may be set up to attract and trap adult beetles.

Coreid bug (*Paradasynus rostratus*)

Symptoms

The attacked buttons become deformed with characteristic crevices on the husk below the perianth with gum exudations and the tender nuts become barren.

Management

Apply 0.1 per cent carbaryl on the newly opened inflorescence after the receptive phase of the female flowers and spray the entire crown excluding the leaves and older bunches.

Note: The insecticide may be applied according to the severity of infection in a need-based manner.

Coconut eriophyid mite

(*Aceria [Eriophyes] guerreronis*)

Coconut eriophyid mite, a recently introduced pest is spreading at an alarming rate in Kerala. It is a microscopic worm like mite infesting young buttons colonizing under the perianth.

Symptoms

The earliest symptoms on 2-3 months old button is pale yellow triangular patches seen below the perianth. Later, these patches become brown. Severely affected buttons may fall. As the buttons grow, brown patches lead to black necrotic lesions with longitudinal fissures on the husk. Uneven growth results in distortion and stunting of nuts leading to reduction in copra yield. In severe cases, the losses are compounded because the quality of fibre is reduced and distorted nuts increase the labour requirements for dehusking.

Management

1. Collect and destroy all the fallen buttons of the affected palm.

2. Apply 2 per cent neem oil + garlic emulsion or commercial neem formulation azadirachtin 0.004 per cent (Neemazal T/S 1 per cent @ 4 ml per litre of water) or micronized wettable sulphur 0.4 per cent in the crown on young bunches. When rocker sprayer is used 1.0 to 1.5 litres of spray fluid per palm is required. If a hand sprayer is used, the spray solution required may be about 500 to 750 ml. Spraying has to be done on second to seventh bunches from top avoiding unpollinated inflorescence. Care should be taken to see that spray fluid reaches the perianth region of third, fourth and fifth bunches since these bunches harbour maximum number of mites. Three rounds of spraying are recommended in a year viz., March-April before the onset of southwest monsoon, in August-September during the dry spell between the southwest and northeast monsoons and in December-January after the northeast monsoon so that all the emerging bunches in the vulnerable stage receive one round of spraying. Rational rotation of the above pesticides may be adopted to avoid chances of resistance.

Preparation of neem oil + garlic emulsion (2 per cent)

To prepare 10 litres of 2 per cent neem oil + garlic emulsion, 200 ml neem oil, 200 g garlic and 50 g ordinary bar soap are required. Slice the bar soap and dissolve in 500 ml lukewarm water. Grind 200 g of garlic and take the

extract in 300 ml of water. Pour the 500 ml soap solution in 200 ml neem oil slowly and stir vigorously to get a good emulsion. Mix the garlic extract in the neem oil + soap emulsion. Dilute this 1 litre stock solution by adding 9 litres of water to get 10 litres of 2 per cent neem oil + garlic emulsion.

As per the recommendation of the National Level Steering Committee, a holistic approach has to be adopted in the management of the coconut eriophyid mite. Hence, in addition to the plant protection measures mentioned above, the following measures can be adopted:

1. Improving nutrient status by applying organic manure at the rate of 50 kg and neem cake 5 kg per palm per year. Also apply the recommended dose of fertilizers in two split applications.
2. Growing compatible intercrops / mixed crops.
3. Providing adequate irrigation.

Mealy bug

Symptoms

Mealy bugs infest the unopened heartleaf and inflorescence. As a result, the leaves become highly stunted, suppressed, deformed and present a crinkled appearance. It is often confused with the leaf rot symptoms. The affected inflorescences are malformed and do not open. Even if they open, they do not bear nuts.

Button mealy bugs colonize under the perianth lobes of tender nuts. Infested nuts harbouring gravid mealy bugs remain on the spadix, which serve as inoculum for further spread.

Management

Remove and destroy all dried up inflorescence and unproductive buttons. Apply non-residual phosphatic insecticides like dimethoate 0.05 per cent or quinalphos 0.05 per cent at the site of infestation. Neem garlic emulsion 2 per cent applied on infested bunches checks button mealy bugs.

Rodents

Rats damage tender nuts by forming characteristic holes. Shed nuts can be seen at the base of the palm.

Place wax blocks of 0.005 per cent bromadiolone in coconut crown of the infested palms at 3 to 4 days interval till the bait is no more consumed.

Diseases

***Phytophthora* diseases**

Phytophthora palmivora has been found to affect seedlings and adult palms causing bud rot and immature nut fall commonly known as mahali.

Bud rot

Symptoms

Palms of all age are liable to be attacked but normally young palms are more susceptible, particularly during monsoon when the temperature is low and humidity is very high. In seedlings, the spear leaf turns pale and comes off with a gentle pull. In adult palms, the first visible symptom is the colour change of the spear, which becomes pale and breaks at the base and hangs down. The tender leaf base and soft tissues of the crown rot into a slimy mass of decayed material emitting a foul smell. The rotting slowly progresses downwards, finally affecting the meristem and killing the palms. This is accompanied by drooping of successive leaves. Even then, nuts that are retained on the palm may grow to maturity. The disease proves fatal if not checked at the early stages, before damage of the bud.

Management

1. In early stages of the disease (when the heartleaf starts withering) cut and remove all affected tissues of the crown. Apply Bordeaux paste and protect it from rain till normal shoot emerges.
2. Burn all disease-affected tissues removed from the palm.
3. Spray 1.0 per cent bordeaux mixture on spindle leaves and crown of disease affected as well as neighbouring palms, as a prophylactic measure. Palms that are sensitive to copper containing fungicides can be protected by mancozeb. Small, perforated sachets containing 2 g of mancozeb may be tied to the top of leaf axil. When it rains, a small quantity of the fungicide is released from the sachets to the leaf base, thus protecting the palm.
4. Adopt control measures for rhinoceros beetle.
5. Provide adequate drainage in gardens.
6. Adopt proper spacing and avoid over crowding in bud rot prone gardens.

Mahali

Symptoms

Shedding of female flowers and immature nuts are the common symptoms of the disease. Lesions appear on the young fruits or buttons near the stalk, which later lead to the decay of the underlying tissues and endosperm.

Management

Spray 1.0 per cent bordeaux mixture or copper oxychloride preparation (0.2 per cent) on the crown of palms, once before the monsoon and once or twice later on at intervals of 40 days.

Root (wilt) disease

Symptoms

The characteristic symptom is the flaccidity of leaflets. Yellowing of older leaves, necrosis of leaflets and deterioration and decay of root system are other salient features of the disease. The leaflets curve inwardly to produce ribbing so that the whole frond develops a cup like appearance. Abnormal shedding of buttons and immature nuts are also noticed.

Management

Coconut root (wilt) is a non-lethal debilitating disease and the affected palms survive for a long period giving a reasonably good yield. The root (wilt) affected palms are susceptible to diseases like leaf rot and pests like rhinoceros beetle and red palm weevil. So there is a chance of confusing the pests and disease symptom with the root (wilt) disease. Negligence on the management aspects aggravates the malady. Efficient management of palms suspected to be affected by coconut root (wilt) disease demands control of all pests and diseases and imparting natural resistance and health to the palms through proper manuring and agronomic practices. A package of management practices for the effective management of root (wilt) disease is given below:

1. Rogue out palms that are affected severely by root (wilt) and yield less than 10 nuts / palm / year and those, which have contracted the disease before flowering. Replant with disease tolerant material / high yielding hybrids (Chandrasankara).
2. Apply fertilizers for coconut palms in average management at the rate of 0.34 kg N, 0.17 kg P₂O₅ and 0.68 kg K₂O / palm / year in the form of urea, rock phosphate and muriate of potash, respectively. For palms under good management, fertilizers may be given @ of 0.5 kg N, 0.32 kg P₂O₅ and 1.2 kg K₂O / palm / year.
3. In addition to the above, apply 50 kg cattle manure or green manure and 1 kg of lime / palm / year. Magnesium may be supplied @ 500 g MgO per palm per year in the Onattukara region (sandy soil) and 100 g MgO in the remaining areas. The cheapest source of MgO is magnesite (MgCO₃). The magnesium in magnesite is acid soluble. Hence it may be preferred in acid soils.
4. Growing green manure crops like sun hemp, sesbania, cowpea and calapagonium in the coconut basin and their incorporation *in situ* is beneficial as the practice reduces the intensity of the root (wilt) and increases the nut yield. The ideal green manure crops for the sandy and alluvial soils are cowpea and sesbania, respectively.
5. Under rainfed conditions, apply fertilizers in two splits, 1/3rd at the time of early southwest monsoon and 2/3rd before the northeast monsoon. Under irrigated conditions apply fertilizers in three equal splits (April-May, August-September and December-January).
6. Apply fertilizers and manures in 10 cm deep circular basins at a radius of 2 m from the bole of the palm.
7. When the crop is grown under the bund and channel system, desilt the channel and strengthen the bunds during summer months.
8. Follow strictly all the prescribed prophylactic measures against leaf rot disease, red palm weevil, rhinoceros beetle etc. so as to ensure that the palms are not debilitated. To maintain the productivity of the palms, prophylactic measures are of great importance.

Leaf rot

Symptoms

The first symptom is the appearance of water-soaked brown lesions in the spear leaves of root-wilt affected palms. Gradually these spots enlarge and coalesce resulting in extensive rotting. As the leaf unfurls, the rotten portions of the lamina dry and get blown off in wind, giving a 'fan' shape to the leaves. Some times, the symptom becomes very acute and the spear fails to unfurl. This disease is a fungal complex initiated predominantly by *Colletotrichum gloeosporioides*, *Exserohilum rostratum* and *Fusarium* spp.

Management

1. Remove the rotten portions from the spear and the two adjacent leaves.
2. Pour 300 ml of fungicidal solution at the base of the spear. This can be prepared by mixing hexaconazole 5 EC 2 ml or mancozeb 3 g in 300 ml water.
3. Treat the top two leaf axils with insecticide preparation. This can be prepared by mixing carbaryl 20 g with 200 g sand.
4. Spray crowns and leaves with 1 per cent bordeaux mixture or 0.2 per cent copper oxychloride formulations or 0.3 per cent mancozeb in January, April-May and September. While spraying, care has to be taken to spray the spindle leaf.

Stem bleeding

Symptoms

Exudation of the reddish brown liquid through the growth cracks mostly at the basal part of the trunk and bleeding patches higher up in the trunk are characteristic symptoms. One or more lesions, lying close by, may coalesce to form large patches. The liquid that oozes out dries up and turns black. The tissues beneath the bleeding points decay and become yellowish. The lesions spread upwards as the disease progresses. In advanced stages, the leaf size reduces leading to reduction in crown size. The rate of leaf production slows down. The production of bunches is affected and nut shedding takes place. The trunk gradually tapers towards the apex. The progress of the disease is faster during July to November.

Causal organism

The fungus, *Thielaviopsis paradoxa* is the causal agent. Growth cracks on the trunk, severe summer followed by heavy down pour, water stagnation, imbalance in nutrition, excess salinity and stress can act as predisposing / aggravating factors.

Management

1. Chisel out completely the affected tissues and paint the wound with tridemorph 5 per cent. Apply coal tar after 1-2 days.
2. Destroy the chiseled materials by burning. Avoid any mechanical injury to trunk.
3. Apply neem cake @ 5 kg per palm in the basin along with other organics.
4. Root feed with tridemorph 5 per cent, thrice a year during April-May, September-October and January-February to prevent further spread of lesions.
5. Apply tridemorph @ 25 ml in 25 litre of water as soil drenching once in four months.

Grey blight

This is caused by the fungus *Pestalotia palmarum*.

Symptoms

Symptoms appear in the mature leaves of the outer whorl as yellow specks encircled by a greyish band which later become greyish white. The spots coalesce into irregular necrotic patches causing extensive leaf blight. In advanced stages, the tips and margins of the leaflets dry and shrivel giving a burnt appearance.

Management

Remove severely affected older leaves and burn. Spray the trees with 1 per cent Bordeaux mixture or propiconazole 0.025 per cent.

Tanjore wilt**Symptoms**

This disease is of recent occurrence in many parts of Kerala, especially in the districts of Palakkad, Malappuram, Thrissur, Kollam, Thiruvananthapuram and Wayanad. Middle aged palms were seen fatally affected. The characteristic symptom of the disease is the rotting of the basal portion of the stem. The bark turns brittle and often gets peeled off in flakes, leaving open cracks and crevices. The internal tissues are discoloured and disintegrated, emitting a bad smell. Mild bleeding occurs on the basal region. The tissues on the bleeding spots are soft to touch. Extensive damage of the root system following root rotting has been observed. Ultimately the palm dies off.

Management

1. Apply organic manure @ 50 kg / palm.
2. Apply neem cake @ 5 kg / palm / year.
3. Reduce fertilizer application to one-fourth of the recommended dose.
4. Drench the basin with 40 litres of 1 per cent Bordeaux mixture or tridemorph 0.1 per cent or any other copper fungicide to soak soil up to 15 cm depth at quarterly intervals.
5. Root feed with tridemorph 2 ml mixed with 100 ml water at quarterly intervals.
6. Avoid flood irrigation in order to prevent the possible spread of the pathogen through soil.
7. Isolate the affected palm from the healthy ones by digging a trench of size 1 m deep and 50 cm wide, 1.5 m away from the bole of the infected palm.
8. Avoid growing leguminous crops in and around the garden.

Safe storage of copra

Copra obtained from commonly cultivated varieties / cultivars is attacked by various insect pests in store. Among these ham beetle, *Necrobia rufipes* and saw toothed grain beetle, *Oryzophilus surinamensis* are of major importance, which can cause more than 15 per cent loss to copra when stored for more than six months.

Following precautions are to be taken for the safe storage of copra for more than three months:

- (1) Dry the produce to 4 per cent moisture content.
- (2) Avoid heap storage, which causes maximum damage.
- (3) Store copra in netted polythene bags or gunny bags.

Shell fired copra dryer

A new type of dryer working on indirect heating and natural convection principles using coconut shell as fuel has been designed and developed by CPCRI. The overall dimension of the dryer is 2.25m length, 1.5m breadth and 1.5m height. The capacity of the dryer is 1000-1200 nuts per batch. The dryer has two heating chambers which are arranged in parallel. Specially developed rolling in type of fuel trays are used for burning the fuel. The dryer consists of a drying chamber, a burning chamber, a plenum chamber and ventilation holes. Each full tray produces heat for 6 hours, with a temperature of about 80-82⁰ C. Generally after 6 hours, when the temperature drops below 60⁰C, the fuel trays are removed from the dryer, cleaned and reloaded with fuel, refired and replaced into respective burning chambers. About 4 loads of fuel are required with a capacity of 80 shells per tray to dry the copra to about 6.25 per cent moisture content. The total drying time is 24 hours. The cost of the dryer is approx Rs.35,000/-. Farmers and entrepreneurs interested to purchase this dryer can contact, Agricultural Technology Information Centre (ATIC), CPCRI, Kasargod, for further details.

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GROUNDNUT (*Arachis hypogaea*)

Groundnut can be cultivated as a floor crop in coconut gardens, as an intercrop with tapioca and as a catch crop after second crop paddy with irrigation. The crop is grown best in sandy loam and loam soils.

Season

Rainfed : May-June to September-October

Irrigated : January to May

Varieties

	<i>Duration (days)</i>
TMV-2 (bunch type)	110
TMV-7 (bunch type)	110
TG-3 (bunch type)	100-110
TG-14 (bunch type)	105-115
Spanish Improved	100-110
Sneha (bunch type)	Early
Snigtha (bunch type)	Early

Note: TG-3, TMV-2 and TMV-7 are recommended as intercrops in coconut gardens.

Seeds and sowing

Pure crop	100kg kernels/ha ⁻¹
Intercrop in coconut	80kg kernels/ha ⁻¹
Intercrop in tapioca	40-50kg kernels/ha ⁻¹

Plough the field three or four times into a fine tilth. Sow the seeds by dibbling in ploughed furrows at a spacing of 15 cm x 15 cm. For seed treatment the rhizobial culture is recommended.

Manuring

Cattle manure or compost	2 t/ha ⁻¹
Lime	1-1.5 t/ha ⁻¹
Fertilizers	10:75:75
N:P ₂ O ₅ :K ₂ O	kg/ha ⁻¹

Apply entire quantity of cattle manure or compost and recommended quantity of fertilizers as basal dressing and incorporate well into the soil. Apply lime at the time of flowering of the crop and mix with the soil by light hoeing or raking.

Irrigation and interculture

Irrigate the crop once in 7 days. Weed the crop 10-15 days after germination of seed by light hoeing. Give another light hoeing or raking at the time of application of lime. Do not disturb the soil after 45 days of sowing.

Plant protection

Red hairy caterpillar, termites and leaf miners are the major pests of groundnut. Against leaf miner apply any of the organophosphorus insecticides with contact action. Spraying carbaryl 0.15 per cent controls red hairy caterpillars. Against ants / earwigs/ termites damaging the pods, apply carbaryl 10 per cent DP in soil at the time of seeding.

Tikka leaf spot disease infects both the rainfed and irrigated crops. Prophylactic spraying with Bordeaux mixture 1.0 per cent before flowering will control the disease.

Harvesting

The crop will be ready for harvest when the leaves start yellowing and begin to dry up. Development of brown colour inside the pods also indicates maturity of the crop.

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OIL PALM (*Elaeis guineensis*)

Oil palm grows best in areas with a mean maximum temperature of 30-32°C and on an average of at least five hours of sunlight. It can be grown in areas, which receive well-distributed annual rainfall of 200 cm or more. However, it can tolerate two to four months of dry spell. The oil palm grows on wide range of tropical soils. The adult palms can withstand occasional waterlogging, but frequently waterlogged, extremely sandy and hard lateritic soils should be avoided.

Varieties

The only variety recommended for commercial cultivation is Tenera, which is a hybrid between Dura and Pisifera.

Nursery practices

The fruits are separated from the bunch and seeds are extracted by scraping off the exocarp and mesocarp with a knife, or by retting in water. The seeds are then dried by spreading them on concrete or wooden floors under shade for two days. Such seeds can be stored for 3-9 months at about 27°C without much reduction in viability.

Seeds are soaked in water for five days, changing the water daily. Thereafter, the seeds are spread out to dry for 24 hours. The dried seeds are put in polythene bags and placed in germinator maintained at a temperature of 40°C. After 80 days, the seeds are removed from polythene bags, soaked in water for 5 days changing the water daily and dried in the shade for two hours. The seeds are then put back into bags and kept in a cool place in order to maintain the moisture content. Germination commences in about 10-12 days. The percentage of germination obtainable by this method is 90-95.

Raising nursery

Polybags (preferably black) of 400-500 gauge measuring 40 cm x 35 cm are used. The bags are filled with topsoil and compost and are arranged at a spacing of 45 cm x 45 cm and one sprouted seed is dibbled per bag. A good mulching during summer is desirable. Watering the seedlings weekly thrice is recommended. A fertilizer mixture containing 15 g N, 15 g P₂O₅ and 6 g K₂O at the rate of 8 g in five litres of water for 100 seedlings may be applied when the seedlings are two months and eight months old.

Planting

Oil palm is planted in the main field in triangular system at spacing of 9 m accommodating 140 palms per ha. Planting is preferably done at the onset of monsoon during May-June. The polythene bag is torn open and the entire ball of earth is buried in the pit (50 cm x 50 cm x 50 cm) and levelled.

Manuring

The following fertilizer schedule is considered satisfactory for oil palm.

N:P ₂ O ₅ :K ₂ O	g/palm/year
First year	400:200:400
Second year	800:400:800
Third year onwards	1200:600:1200

Mg application is necessary only if deficiency symptoms are noticed. Fertilizers are preferably applied in two equal split doses (May and September), within 2 m diameter around the palm and forked in. Supply of sufficient quantities of green leaf or compost is advantageous, especially where the soil is poor in organic matter.

Leaf pruning

Dead and diseased leaves and all inflorescences should be cut off regularly upto three years after planting. When the palms are yielding, judicious pruning to retain about 40 leaves on the crown is advocated. It is necessary to remove some of the leaves while harvesting. In such cases, care should be taken to avoid over pruning. In addition, all dead and excess leaves should be cut off and crown cleaned at least once in a year, usually during the dry season.

Pollination

Oil palm is a cross-pollinated crop. Assisted pollination is done to ensure fertilization of all female flowers. However, this is not necessary if the pollination weevil *Elaeobius kamerunicus* is introduced in the plantation. They

congregate and multiply on male inflorescence during flower opening. The weevils also visit the female flowers and pollinate them effectively.

Harvesting

First harvest can be taken 3.5 to 4 years after planting. When a few ripe fruits are loose / fall off, the bunch is ready for harvesting. Processing over-ripe fruits reduces quantity and quality of oil.

A chisel is used for harvesting bunches from young palms. The stalk of the bunch is struck hard with the chisel to cut off and push the bunch out. When the palms become taller (from 10 year onwards) a harvesting hook has to be used. When the palms are too tall, it is necessary to climb the palms for harvesting.

Plant protection

Pests

Rhinoceros beetle

The pest causes severe damage to emerging fronds and spindle. The adult beetle feeds on the softer tissue of the rachis, resulting in snapping off of the fronds and spears at the feeding sites. Field sanitation and elimination of breeding sites are essential components of the pest management operation. This pest can be suppressed by using the virus *Baculovirus oryctes*.

Red palm weevil

This is a major pest of oil palm in India. These weevils lay their eggs at the cut end of petioles or other wounds. The emerging larvae tunnel into the crown and feed on the growing tissues. Palms infested by red palm weevil start wilting and leaves show gradually increasing chlorosis and fracture in strong winds. If detected early, treatment of affected palm with 0.2 per cent carbaryl would save the palms.

Birds

Many birds such as the forest crow, the house crow and the common Indian myna cause severe damage to oil palm fruit bunches. These birds feed on the mesocarp of the oil palm fruits. The damage can be minimized by scaring the birds and covering the ripe bunch with wire net, 150 days after fruit set.

Diseases

Anthrachnose

This disease occurs in the nursery. It is recognized by regular or irregular brown to black leaf blotches surrounded by yellow haloes, which develop along the margin, centre or tip of the leaves. It causes heavy seedling loss. The disease can be controlled by spraying mancozeb or captan @ 200 g/100 litres of water. Copper fungicides should not be used because of the extreme susceptibility of oil palm seedlings to copper burn (scorching).

Spear rot

This is noticed to affect oil palms of all ages. The incidence is less than one per cent. Yellowing starts from tip of the innermost whorl of leaves. Small lesions occur at the distal portions of spear and rotting extends downwards. As the disease advances, new leaves become rudimentary and show rotting. General decline in vigour and production is then noticed. Occurrence of spear rot without yellowing has also been noticed. Distinguishable marginal yellowing of leaflets and sudden drying of leaves showing yellowing are other symptoms. Roguing of all the affected palms may be adopted to prevent further spread of the disease. In early stages of the disease, the affected portions of leaves may be removed and burnt.

Bunch failure

Sparse or no fruit set followed by complete drying or rotting of the affected bunches are the typical symptoms. The extent of incidence can be up to 20 per cent. This malady is generally attributed to excess pruning, mutual shading, underpollination, moisture stress and unhygienic conditions. The situation can be improved by assisted pollination as well as by adopting hygienic measures like removal of infected bunches and dry male inflorescence.

Processing

For mature plantations not exceeding 40 ha, a hand-operated hydraulic press will be enough for extraction of oil. In the case of large-scale plantations, the hydraulic press will not be economical and as such, mechanically driven oil mills have to be established. The fruit bunches brought to the factory are first quartered by means of a chisel. They are then sterilized in steam or boiling water for 30-60 minutes. The objective of this process is to inactivate the fat splitting enzymes, which are present in the fruit, which may raise the free fatty acid content of the oil and also to soften the fruits for easy pounding. The sterilized fruits are stripped off from the bunch and then pounded. The pounded fruit mass is then reheated and squeezed using a hydraulic press. It is then boiled in a clarification drum where the sludge will deposit and pure oil float over the water. The oil is then drained out.

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Sesame grows in well-drained, sandy loam soils. Coarse sandy soils and soils of alkaline and saline nature are not suited for the crop.

Season

Lowland paddy field: December_April (third crop)

Uplands: August_December

Varieties

For upland cultivation use varieties with long duration of 100-110 days and for low land, use varieties with duration of 80-99 days.

Table 22. Sesame varieties

Kayamkulam-1	Suitable for lowlands of Onattukara
Kayamkulam-2 (Thilothama)	Suitable for rice fallows in Onattukara, resistant to leaf spot disease
ACV-1 (Soma)	Pure line selection for summer fallows of Onattukara
ACV-2 (Surya)	Pure line selection suited for uplands
ACV-3 (Thilak)	Pure line selection suited to summer fallows of Onattukara
Thilathara (CST 785 x B14)	Suitable for the summer rice fallows of Onattukara, oil 51.5%, duration 78 days
OMT-1165	Suited to uplands of Onattukara (rabi season), oil 50.5%
Thilarani	Suited to summer rice fallows of Onattukara

Preparation of land and sowing

Prepare the soil into a fine tilth by ploughing 2-4 times and breaking the clods. Seed rate is 4-5 kg ha⁻¹. Broadcast seeds evenly, preferably mixed with sand 2-3 times its volume, to ensure uniform coverage. Work with harrow, followed by pressing with wooden plank so as to cover the seed in the soil.

Manuring

Apply manures and fertilizers at the following rates.

Cattle manure / compost	5 t ha ⁻¹
N:P ₂ O ₅ :K ₂ O	30:15:30 kg ha ⁻¹

Apply cattle manure/compost as basal dressing and incorporate into the soil along with last ploughing. Apply fertilizers as basal dose when there is enough moisture in the soil. Urea is preferable to ammonium sulphate. Nitrogen may be applied in split doses, 75 per cent as basal and the balance as foliar spray at 3 per cent concentration, 20-35 days after sowing keeping the discharge rate at 500 l ha⁻¹.

Interculture

Aftercultivation of the crop may be done twice, first at 15 days and the next 25-35 days after sowing. When the plants are about 15 cm in height, thin the crop so as to give a spacing of 15-25 cm between plants.

Irrigation

Usually the crop is grown under rainfed conditions. When facilities are available, the crop may be irrigated to field capacity after thinning operation and thereafter at 15-20 days interval. Stop irrigation just before the pods begin to mature.

Surface irrigation at 3 cm depth during the critical stages, viz., 4-5 leaves, branching, flowering and pod formation will increase the yield by 35-52 per cent. Two irrigations of 3 cm depth each in the vegetative phase (4-5 leaf stage or branching) and in reproductive phase (at flowering or pod formation) are the best, registering maximum yield and water use efficiency. In the case of single irrigation, it can be best given in the reproductive phase. In the tail end fields in command area, best use of the sparingly available water can be made for augmenting sesame production.

Plant protection

For control of leaf and pod caterpillar, remove affected leaves and shoots and dust with carbaryl 10 per cent. Azadirachtin 0.03 per cent at 5 ml per litre spray at 7th and 20th DAS and thereafter need based application can manage the incidence of leaf and pod caterpillar, pod borer infestation and phyllody incidence. For control of gall fly, give preventive spray with 0.2 per cent carbaryl.

For control of leaf curl disease, remove and destroy disease affected sesame plants as well as the diseased collateral hosts like chilli, tomato and zinnia.

Remove plants affected with phyllody and destroy them. Do not use seeds from affected plants for sowing.

Harvesting

Harvest the crop, when the capsules turn yellowish by pulling out the plants. Harvest during the morning hours. Cut the root portion and stack the plants in bundles for 3-4 days when the leaves will fall off. Spread in the sun and beat with sticks to break open the capsules. Repeat this for 3 days. Preserve seeds collected during the first day for seed purposes. Clean and dry in sun for about 7 days before storing.

Storage of seeds

By keeping sesame seeds in polybags, tin bins, wooden receptacles or in earthen pots, the viability can be maintained for about one year. Admixture of seeds with ash will drastically reduce germination.

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